

Primal–Dual Completion Certificates for Fixed-Shift Primitive Entry Kernels: Exact Quotient Gauges, Obstruction Witnesses, and Perron Aggregation

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Abstract

TPC–73 isolated an analytic completion gate on the primitive cofactor plane, and TPC–74 attached that plane to every off-diagonal entry of the normalized missing-native Gram. We make the remaining completion infimum exactly decidable. On every finite rectangle, a linear operator A_D has norm $\|A_DF\|_1$ equal to the full low-diagonal plaquette cost plus exact high-diagonal tail. If R is primitive restriction, we prove attainment and the strong complex dual identity

$$\min_{RF=k} \|A_DF\|_1 = \max_{\substack{A_D^* z = R^* \lambda \\ \|z\|_\infty \leq 1}} \operatorname{Re}\langle \lambda, k \rangle.$$

The gauge is a quotient seminorm with nullspace $R(\ker A_D)$; a low-cost completion and a dual obstruction witness are exact alternatives, and the physical Möbius functional is one feasible dual direction.

For a finite Hermitian row family, let γ_{mn} be the optimal cost of the fixed post- q direct-entry kernel. When admissible completions factor over unordered row pairs, we prove the exact matrix-optimization identity

$$\inf_{\mathbf{F}} \rho(P(\mathbf{F})) = \rho(Q^\gamma), \quad Q_{mn}^\gamma = \gamma_{mn}.$$

Thus Hermitian reversal alone creates no joint optimization gain. We also derive a strong dual for fixed weighted-Schur aggregation under an arbitrary declared affine coupling rule at fixed finite stacked data; this is where shared q -compatibility creates a genuine joint gate. Uniformity over a growing coefficient class remains a separate minimax problem.

Finally, every completion majorant is bounded below by the absolute-entry Perron radius of the gauged direct-entry matrix. A zero-diagonal symmetric Hadamard construction has $\|H\| = O(N^{-1/2})$ while $\rho(|H|) \rightarrow 1$, proving that failure of all absolute-value completion certificates need not mean failure of signed operator cancellation. These are exact certificate/obstruction results, not a physical saving; no fixed-2, parity-breaking, prime-pair, or twin-prime conclusion is asserted.

Keywords: Möbius function; analytic completion; convex duality; quotient seminorm; obstruction witness; fixed shift; Hermitian Gram matrix; Perron majorant.

Convention. Every physical specialization retains one fixed nonzero shift h_0 , literal coefficients, complete native output keys, the actual active support, and raw atomic normalization. No statement is specialized to $h_0 = 2$.

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1 Introduction

The primitive-plane program has reached a precise optimization boundary. The divisor lift of TPC–71 [3] converts a coprime two-variable Möbius sum into diagonal slices. TPC–73 [4] proves that the

support-faithful physical kernel has no auxiliary slices with $d \geq 2$, and that every nonprimitive analytic completion preserves the primitive physical functional. Its complete certificate cost is nevertheless a coupled sum of low-diagonal plaquette variation and a high-diagonal tail. TPC-74 [5] then constructs the missing interface for the gauged perturbation $\tilde{H} = U_d H U_d$:

$$\tilde{H}_{mn} = \sum_{(u,v)=1} \mu(u)\mu(v)L_{mn}^0(u,v)$$

for every actual off-diagonal row pair of the normalized missing-native Gram. A completion certificate can therefore be asked entry by entry and aggregated into a nonnegative Hermitian majorant.

Three logical gaps remain if the completion infimum is left only as a definition.

- (i) A failed search for a smooth or q -compatible completion is not a proof that no cheaper completion exists.
- (ii) Separate entrywise optimizers need not define one admissible rule when a fixed finite stacked family is required to share parameters across row pairs.
- (iii) Even an exact lower obstruction for every absolute-value completion majorant may obstruct only that certificate architecture, not the signed matrix itself.

The purpose of this paper is to close all three logical gaps at the finite-dimensional level.

1.1 Main results

Fix a finite rectangle and a diagonal cutoff D . The full TPC-73 cost is a finite sum of absolute values of linear functionals of the completion. We stack those functionals, with their literal weights, into one operator A_D . If R is restriction to the primitive lattice, then

$$\mathcal{C}_D(F) = \|A_D F\|_1, \quad \gamma_D(k) = \min_{RF=k} \|A_D F\|_1.$$

Our first result is an exact strong dual over the underlying real spaces:

$$\gamma_D(k) = \max \{ \operatorname{Re} \langle \lambda, k \rangle : A_D^* z = R^* \lambda, \|z\|_\infty \leq 1 \}.$$

The formula treats complex literal coefficients without discarding their phases. It also proves attainment on both sides. Any feasible (λ, z) gives a rigorously checkable lower obstruction, while an optimal completion gives a matching upper certificate.

The second result identifies the geometry of the optimum. The map $k \mapsto \gamma_D(k)$ is a seminorm on the primitive data space, with

$$\ker \gamma_D = R(\ker A_D).$$

It is therefore a norm on the corresponding quotient. This removes a common ambiguity: zero completion cost is possible exactly when the prescribed primitive datum is the trace of a zero-analysis completion, not merely when one chosen interpolation happens to have small variation.

The third result treats the complete direct-entry family. For independent unordered row-pair completion classes, define the symmetric matrix

$$Q_{mn}^\gamma = \begin{cases} \gamma_{mn}, & m \neq n, \\ 0, & m = n. \end{cases}$$

Then

$$\|H\| = \|\tilde{H}\| \leq \rho(|\tilde{H}|) \leq \rho(Q^\gamma) = \inf_{\mathbf{F}} \rho(P(\mathbf{F})).$$

The last equality is exact. Each edge minimum can be attained simultaneously, and Perron monotonicity leaves no hidden joint gain. Hence a genuine joint gate cannot be attributed to Hermitian reversal alone. It appears only when one declares additional cross-edge constraints, such as a shared q -compatible rule or a finite stacked-data parametrization. We give the exact primal and dual for every such finite affine coupled rule; uniformity over a growing coefficient class remains a separate minimax problem.

1.2 What this advances

The paper converts an open-ended search problem into an exact decision problem. At a declared finite scale and within a declared completion class, exactly one numerical value controls both sides:

$$\text{best completion cost} = \text{strongest dual obstruction.}$$

This is a real theoretical advance at the interface level. It does not show that the value is small on the growing physical carrier. Nor does it identify a classical Mertens estimate sufficient for that smallness.

The absolute-entry Perron floor makes the limitation sharper. We give a deterministic symmetric family for which the signed operator norm tends to zero but the absolute-entry spectral radius tends to one. Thus a large dual obstruction can close the entire Abel–completion–Perron route while a different signed-moment route remains viable.

1.3 Organization

[Section 2](#) writes the divisor–variation cost as one finite linear analysis norm. [Section 3](#) proves strong duality and attainment. [Section 4](#) develops the quotient geometry and physical witness. [Section 5](#) proves the exact independent-edge Perron theorem. [Section 6](#) handles shared affine rules. [Section 7](#) records the necessary non-substitution laws. [Section 8](#) returns to the literal fixed-shift carrier, and [section 9](#) gives the claim ledger and next decision.

2 The finite completion problem in linear normal form

2.1 Real pairings on complex spaces

Every complex vector space below is regarded as a finite-dimensional real space when convex duality is used. For vectors of the same dimension set

$$[x, y]_{\mathbb{R}} := \operatorname{Re} \langle x, y \rangle = \operatorname{Re} \sum_j \overline{x_j} y_j.$$

Adjoints are characterized by

$$[Ax, z]_{\mathbb{R}} = [x, A^* z]_{\mathbb{R}}.$$

This convention is essential when a Hermitian reversal contains complex conjugation: conjugation is real-linear but not complex-linear.

2.2 Primitive restriction

Fix positive integers U, V , put

$$B = [1, U] \times [1, V], \quad P = B \cap \{(u, v) : (u, v) = 1\},$$

and let

$$\mathcal{X} = \mathbb{C}^B, \quad \mathcal{K} = \mathbb{C}^P.$$

The restriction map

$$R : \mathcal{X} \longrightarrow \mathcal{K}, \quad (RF)(u, v) = F(u, v) \quad ((u, v) \in P)$$

is surjective. For prescribed primitive data $k \in \mathcal{K}$, its full analytic completion class is the affine space

$$\operatorname{Ext}_B(k) = \{F \in \mathcal{X} : RF = k\}.$$

Zeros at absent primitive rays are part of k ; only nonprimitive sites are free.

2.3 The analysis operator

For $d \geq 1$, define

$$M_d(x) = \sum_{\substack{r \leq x \\ (r,d)=1}} \mu(r), \quad F^{[d]}(x, y) = F(dx, dy),$$

using zero extension outside B . Let

$$\Delta_{12}G(x, y) = G(x, y) - G(x+1, y) - G(x, y+1) + G(x+1, y+1).$$

Fix $D \geq 1$. We use two finite output index sets:

$$J_{\text{low}} = \{(d, x, y) \in \mathbb{N}^3 : 1 \leq d \leq D, dx \leq U, dy \leq V\},$$

$$J_{\text{tail}} = \{(d, x, y) \in \mathbb{N}^3 : d > D, dx \leq U, dy \leq V\}.$$

Indices carrying zero weights may be retained or deleted.

Definition 2.1 (Divisor–variation analysis operator). Let

$$\mathcal{Y}_D = \mathbb{C}^{J_{\text{low}}} \oplus \mathbb{C}^{J_{\text{tail}}}.$$

Define $A_D : \mathcal{X} \rightarrow \mathcal{Y}_D$ by

$$(A_DF)_{d,x,y}^{\text{low}} = |\mu(d)| |M_d(x)M_d(y)| \Delta_{12}F^{[d]}(x, y), \quad (1)$$

$$(A_DF)_{d,x,y}^{\text{tail}} = |\mu(d)\mu(dx)\mu(dy)| F(dx, dy). \quad (2)$$

Proposition 2.2 (Exact normal form). *For every $F \in \mathcal{X}$,*

$$\boxed{\mathcal{C}_D(F) = \|A_DF\|_1}, \quad (3)$$

where $\mathcal{C}_D$ is the complete TPC-73 low-diagonal weighted plaquette cost plus its exact high-diagonal tail.

Proof. The ℓ^1 -sum of the coordinates in (1) is

$$\sum_{d \leq D} |\mu(d)| \sum_{x,y \geq 1} |M_d(x)M_d(y)| |\Delta_{12}F^{[d]}(x, y)|.$$

The ℓ^1 -sum of (2) is

$$\sum_{d > D} |\mu(d)| \sum_{x,y \geq 1} |\mu(dx)\mu(dy)F(dx, dy)|.$$

All sums are finite and their sum is precisely $\mathcal{C}_D(F)$. $\square$

Remark 2.3 (Literal weights are not envelopes). The coordinates of A_D contain the exact restricted Mertens weights, the exact support rectangle, and the literal completion values. Replacing them by maxima, support counts, or a smooth ambient box defines a different and generally coarser operator.

Definition 2.4 (Optimal completion gauge). For $k \in \mathcal{K}$, set

$$\gamma_D(k) := \inf_{F \in \text{Ext}_B(k)} \mathcal{C}_D(F) = \inf_{RF=k} \|A_DF\|_1. \quad (4)$$

Because R is surjective, the feasible set is never empty. We retain the more general range formulation in the dual theorem so that later affine coupled-rule systems can be included without changing notation.

3 Exact primal–dual completion theorem

We first state the theorem for arbitrary finite spaces. This separates the convex fact from the arithmetic realization of A_D .

Theorem 3.1 (Strong completion duality). *Let $\mathcal{X}, \mathcal{Y}, \mathcal{K}$ be finite-dimensional complex spaces, regarded as real spaces with the pairing $[\cdot, \cdot]_{\mathbb{R}}$. Let*

$$A : \mathcal{X} \rightarrow \mathcal{Y}, \quad R : \mathcal{X} \rightarrow \mathcal{K}$$

be complex-linear maps, and define

$$\gamma(k) = \inf\{\|AF\|_1 : RF = k\},$$

with value $+\infty$ when the constraint is infeasible. Then

$$\boxed{\gamma(k) = \sup_{\substack{A^*z = R^*\lambda \\ \|z\|_{\infty} \leq 1}} [\lambda, k]_{\mathbb{R}}.} \quad (5)$$

If $k \in \text{Ran } R$, both the primal minimum and the dual maximum are attained. If $k \notin \text{Ran } R$, the dual is unbounded above.

Proof. The real Fenchel conjugate of the complex ℓ^1 -norm is

$$(\|\cdot\|_1)^*(z) = \begin{cases} 0, & \|z\|_{\infty} \leq 1, \\ +\infty, & \|z\|_{\infty} > 1. \end{cases}$$

Indeed, each complex coordinate is a two-dimensional Euclidean block and

$$\sup_y \{[z, y]_{\mathbb{R}} - |y|\}$$

is finite exactly when $|z| \leq 1$.

For the equality constraint use the Lagrangian

$$\mathcal{L}(F, \lambda) = \|AF\|_1 + [\lambda, k - RF]_{\mathbb{R}}.$$

The infimum over F is finite precisely when there is $z \in \mathcal{Y}$ satisfying

$$\|z\|_{\infty} \leq 1, \quad A^*z = R^*\lambda.$$

In that case the infimum equals $[\lambda, k]_{\mathbb{R}}$; otherwise it is $-\infty$. This gives the displayed dual. Finite-dimensional Fenchel–Rockafellar duality, applied on the nonempty affine space $\{F : RF = k\}$, gives equality whenever $k \in \text{Ran } R$ [2].

For completeness, primal attainment does not require coercivity on all of $\mathcal{X}$. Quotient the feasible direction space $\ker R$ by $\ker A \cap \ker R$, choose a linear complement, and restrict there. The induced map into $\text{Ran } A$ is injective, so its norm controls the complement variable. A minimizing sequence is therefore bounded modulo the cost-free direction, and has a convergent representative.

For dual attainment, restrict λ to $(\ker R^*)^{\perp}$. On that subspace R^* is injective and $A^*z = R^*\lambda$, $\|z\|_{\infty} \leq 1$, bounds λ . The resulting feasible set is closed and bounded, hence compact. Adding a vector in $\ker R^*$ does not change the objective for $k \in \text{Ran } R$.

If $k \notin \text{Ran } R$, choose $\lambda_0 \in (\text{Ran } R)^{\perp}$ with $[\lambda_0, k]_{\mathbb{R}} > 0$. Then $R^*\lambda_0 = 0 = A^*0$, so $(t\lambda_0, 0)$ is dual feasible for every $t > 0$, and its objective tends to $+\infty$. $\square$

Corollary 3.2 (Exact certificate/obstruction alternative). *Suppose $k \in \text{Ran } R$ and $\varepsilon \geq 0$. Exactly one of the following strict alternatives can occur:*

- (a) *there is F with $RF = k$ and $\|AF\|_1 \leq \varepsilon$;*
- (b) *there are λ, z with $A^*z = R^*\lambda$, $\|z\|_{\infty} \leq 1$, and $[\lambda, k]_{\mathbb{R}} > \varepsilon$.*

If the optimum equals ε , the primal and dual optimizers meet at equality rather than satisfying the strict second alternative.

Proof. Both optima equal $\gamma(k)$ by [theorem 3.1](#). Compare this common value with ε . $\square$

Remark 3.3 (What is checkable). A proposed completion is verified by the equality $RF = k$ and a direct evaluation of $\|AF\|_1$. A proposed obstruction is verified by one adjoint equality and one coordinatewise unit-ball inequality. Neither certificate requires trusting an optimizer or a numerical convergence diagnostic.

Corollary 3.4 (KKT certificate). *For feasible k , a primal–dual feasible quadruple (F, λ, z, k) is optimal if and only if*

$$[z, AF]_{\mathbb{R}} = \|AF\|_1. \quad (6)$$

Equivalently, coordinatewise,

$$z_\nu = \frac{(AF)_\nu}{|(AF)_\nu|} \quad \text{when } (AF)_\nu \neq 0, \quad |z_\nu| \leq 1 \quad \text{when } (AF)_\nu = 0.$$

Proof. Primal and dual feasibility give

$$[\lambda, k]_{\mathbb{R}} = [R^* \lambda, F]_{\mathbb{R}} = [A^* z, F]_{\mathbb{R}} = [z, AF]_{\mathbb{R}} \leq \|AF\|_1.$$

Equality is exactly simultaneous optimality by strong duality. Equality in each complex coordinate support inequality gives the stated phase condition. $\square$

Remark 3.5 (LP versus second-order cone form). For real kernels the absolute values give a linear program after the usual positive/negative epigraph split. For complex kernels, each constraint $|z_j| \leq 1$ is a two-dimensional second-order cone. Thus the exact finite problem is an SOCP in general and an LP only in the real specialization. No asymptotic conclusion follows merely from solving either finite program.

Corollary 3.6 (TPC–73 gauge dual). *For the operator A_D in [definition 2.1](#) and primitive restriction R ,*

$$\gamma_D(k) = \max_{\substack{A_D^* z = R^* \lambda \\ \|z\|_\infty \leq 1}} \operatorname{Re} \langle \lambda, k \rangle. \quad (7)$$

Proof. Use [proposition 2.2](#) and [theorem 3.1](#). Here R is surjective, so both optima are finite and attained for every $k \in \mathcal{K}$. $\square$

4 Quotient geometry and arithmetic witnesses

4.1 The exact nullspace

Theorem 4.1 (Completion quotient seminorm). *On $\operatorname{Ran} R$, the optimal completion cost*

$$\gamma(k) = \min_{RF=k} \|AF\|_1$$

is a seminorm and

$$\ker \gamma = R(\ker A). \quad (8)$$

Consequently it induces a norm on

$$\operatorname{Ran} R / R(\ker A).$$

Proof. Positive homogeneity follows by scaling a completion. If $RF_i = k_i$, then $R(F_1 + F_2) = k_1 + k_2$ and

$$\|A(F_1 + F_2)\|_1 \leq \|AF_1\|_1 + \|AF_2\|_1.$$

Taking minima gives the triangle inequality.

If $k = RF$ for some $F \in \ker A$, then $\gamma(k) = 0$. Conversely, if $\gamma(k) = 0$, primal attainment in [theorem 3.1](#) supplies F with $RF = k$ and $\|AF\|_1 = 0$, so $F \in \ker A$. $\square$

Corollary 4.2 (Stability). *For $k, k' \in \text{Ran } R$,*

$$|\gamma(k) - \gamma(k')| \leq \gamma(k - k').$$

Thus the gauge is Lipschitz with constant one in its own quotient norm.

Proof. Apply the triangle inequality twice. □

4.2 The dual unit ball

Define

$$\mathcal{D}_{A,R} = \{\lambda \in \mathcal{K} : \exists z, A^*z = R^*\lambda, \|z\|_\infty \leq 1\}.$$

When R is surjective this set is compact, convex, balanced, and

$$\gamma(k) = \max_{\lambda \in \mathcal{D}_{A,R}} [\lambda, k]_{\mathbb{R}}. \quad (9)$$

It is the unit ball of the dual of the quotient norm in [theorem 4.1](#). A lower obstruction is therefore not a heuristic roughness statistic; it is a separating functional in the exact dual unit ball.

Proposition 4.3 (Zero-cost criterion by annihilators). *For $k \in \text{Ran } R$, the following are equivalent:*

- (i) $\gamma(k) = 0$;
- (ii) $k \in R(\ker A)$;
- (iii) $[\lambda, k]_{\mathbb{R}} = 0$ for every λ satisfying $R^*\lambda \in \text{Ran } A^*$.

Proof. The first two statements are [theorem 4.1](#). In finite dimensions,

$$(R(\ker A))^\perp = \{\lambda : R^*\lambda \in (\ker A)^\perp\} = \{\lambda : R^*\lambda \in \text{Ran } A^*\}.$$

The third statement says exactly that k belongs to the double orthogonal complement of $R(\ker A)$. □

4.3 The physical Möbius direction

For the primitive lattice P , define

$$\sigma(u, v) = \mu(u)\mu(v) \quad ((u, v) \in P).$$

The physical functional is

$$\mathcal{S}_{\text{prim}}(k) = \langle \sigma, k \rangle.$$

Theorem 4.4 (Arithmetic dual witness). *For every $D \geq 1$, there is $z_D^\sigma \in \mathcal{Y}_D$ such that*

$$A_D^* z_D^\sigma = R^* \sigma, \quad \|z_D^\sigma\|_\infty \leq 1. \quad (10)$$

Consequently,

$$|\mathcal{S}_{\text{prim}}(k)| \leq \gamma_D(k). \quad (11)$$

Proof. The exact diagonal divisor lift followed by Abel summation [\[3\]](#) gives, for every completion F ,

$$\begin{aligned} \langle \sigma, RF \rangle &= \sum_{d \leq D} \mu(d) \sum_{x,y} M_d(x) M_d(y) \Delta_{12} F^{[d]}(x, y) \\ &\quad + \sum_{d > D} \mu(d) \sum_{x,y} \mu(dx) \mu(dy) F(dx, dy). \end{aligned}$$

Explicitly, with $\text{sgn}(0) = 0$, take

$$\begin{aligned} (z_D^\sigma)_{d,x,y}^{\text{low}} &= \text{sgn}(\mu(d) M_d(x) M_d(y)), \\ (z_D^\sigma)_{d,x,y}^{\text{tail}} &= \text{sgn}(\mu(d) \mu(dx) \mu(dy)). \end{aligned}$$

Thus every nonzero coordinate is the corresponding real coefficient divided by its absolute value. This defines z_D^σ with coordinate modulus at most one and

$$\langle z_D^\sigma, A_DF \rangle = \langle \sigma, RF \rangle \quad (F \in \mathcal{X}).$$

Hence $A_D^* z_D^\sigma = R^* \sigma$.

For any unit complex number ω , the pair $(\omega\sigma, \omega z_D^\sigma)$ is also dual feasible. Choose ω so that $\operatorname{Re}\langle \omega\sigma, k \rangle = |\langle \sigma, k \rangle|$, and apply [corollary 3.6](#). $\square$

Remark 4.5 (The witness need not be optimal). The arithmetic sign witness recovers the original TPC–73 upper certificate for the physical functional. The dual optimizer may give a strictly larger value than $|\mathcal{S}_{\text{prim}}(k)|$. Such a gap obstructs the chosen absolute-value certificate architecture; it does not prove that the physical signed functional is large.

Corollary 4.6 (Single-atom rigidity). *If $k = ce_{u_0, v_0}$ with $(u_0, v_0) = 1$ and $\mu(u_0)\mu(v_0) \neq 0$, then*

$$\gamma_D(k) \geq |c|.$$

No nonprimitive analytic fill can erase this nonzero physical atom.

Proof. Use [equation \(11\)](#). $\square$

5 Hermitian row-pair families and the exact Perron gauge

5.1 Unordered-edge convention

Let $\mathcal{M}$ be a nonempty finite active row set and

$$\mathcal{E} = \{\{m, n\} : m, n \in \mathcal{M}, m < n\}.$$

For every edge $e = \{m, n\}$, choose one upper-triangle completion rectangle B_e , primitive datum k_e , restriction R_e , cutoff D_e , and analysis operator $A_e = A_{D_e}$. The reversed rectangle is the transpose of B_e , and the lower-triangle completion is defined by

$$F_{nm}(v, u) = \overline{F_{mn}(u, v)}. \quad (12)$$

We assume the inherited TPC–74 reversal compatibility:

$$k_{nm}(v, u) = \overline{k_{mn}(u, v)}, \quad \|A_{nm}F_{nm}\|_1 = \|A_{mn}F_{mn}\|_1.$$

Encoding only $m < n$ avoids treating the real-linear conjugate reversal as a complex-linear constraint.

For an admissible upper-triangle family $\mathbf{F} = (F_e)_{e \in \mathcal{E}}$, define

$$p_e(F_e) = \|A_e F_e\|_1$$

and the symmetric nonnegative matrix

$$P(\mathbf{F})_{mn} = \begin{cases} p_{\{m, n\}}(F_{\{m, n\}}), & m \neq n, \\ 0, & m = n. \end{cases} \quad (13)$$

The single-edge optimal costs are

$$\gamma_{mn} = \min_{R_{mn}F = k_{mn}} \|A_{mn}F\|_1 \quad (m < n),$$

and $\gamma_{nm} = \gamma_{mn}$. Set

$$Q_{mn}^\gamma = \begin{cases} \gamma_{mn}, & m \neq n, \\ 0, & m = n. \end{cases} \quad (14)$$

5.2 No hidden independent-edge gain

Theorem 5.1 (Exact separated Perron completion). *Suppose the admissible family is the product of the unordered-edge completion classes:*

$$R_e F_e = k_e \quad (e \in \mathcal{E}),$$

with no additional cross-edge constraint. Then

$$\boxed{\inf_{\mathbf{F}} \rho(P(\mathbf{F})) = \rho(Q^\gamma)}. \quad (15)$$

The infimum is attained.

Proof. For every admissible family,

$$P(\mathbf{F})_{mn} \geq Q_{mn}^\gamma$$

entrywise. Perron spectral radius is monotone on nonnegative matrices [1], so

$$\rho(P(\mathbf{F})) \geq \rho(Q^\gamma).$$

Each single-edge minimum is attained by [theorem 3.1](#). Because $\mathcal{E}$ is finite and the classes are independent, choose all upper-edge minimizers simultaneously and reconstruct the lower triangle by (12). For that family $P(\mathbf{F}) = Q^\gamma$. $\square$

Remark 5.2 (Where a joint gap can arise). Hermitian reversal alone causes no gap between entrywise and matrix optimization. A genuine joint problem requires an additional shared rule: a common parameter, a common content-lift formula, cross-edge regularity, a uniform cutoff, or a rule fixed over a coefficient class. Those constraints must be written into the primal feasible set.

5.3 The universal absolute-entry floor

Let $\tilde{H}$ be the exact gauged Hermitian zero-diagonal matrix whose entries are the primitive signed functionals of the k_{mn} . If H is the ungauged missing-native perturbation of TPC-74, then

$$\tilde{H} = U_d H U_d, \quad U_d = \text{diag}(\mu(d_m)).$$

Thus $|\tilde{H}| = |H|$ entrywise and $\|\tilde{H}\| = \|H\|$. Explicitly,

$$\tilde{H}_{mn} = \sum_{(u,v)=1} \mu(u)\mu(v)k_{mn}(u,v) \quad (m < n),$$

exactly.

Theorem 5.3 (Perron floor and canonical ceiling). *For every admissible completion family,*

$$\boxed{\|H\| = \|\tilde{H}\| \leq \rho(|\tilde{H}|) \leq \rho(Q^\gamma) \leq \rho(P(\mathbf{F}))}. \quad (16)$$

In particular, if P^0 is the canonical primitive zero-completion matrix, then

$$\|H\| = \|\tilde{H}\| \leq \rho(|\tilde{H}|) \leq \rho(Q^\gamma) = \inf_{\mathbf{F}} \rho(P(\mathbf{F})) \leq \rho(P^0). \quad (17)$$

Proof. [Theorem 4.4](#) gives

$$|\tilde{H}_{mn}| \leq \gamma_{mn} \leq P(\mathbf{F})_{mn}.$$

All three right-hand matrices are symmetric and nonnegative, so Perron monotonicity gives the spectral-radius inequalities. Finally, for arbitrary vectors x, y ,

$$|\langle x, \tilde{H}y \rangle| \leq \langle |x|, |\tilde{H}||y| \rangle \leq \rho(|\tilde{H}|) \|x\|_2 \|y\|_2,$$

which proves $\|H\| = \|\tilde{H}\| \leq \rho(|\tilde{H}|)$. $\square$

Remark 5.4 (Meaning of the floor). No completion whose final step takes entrywise absolute values can beat $\rho(|\tilde{H}|) = \rho(|H|)$. A large floor closes that certificate architecture, not the possibility that signs in H make $\|H\|$ small.

5.4 Fixed weighted-Schur aggregation

Fix $c = (c_m)_{m \in \mathcal{M}} \in (0, \infty)^{\mathcal{M}}$. Define

$$\Theta_c(\mathbf{k}) = \min_{\mathbf{F}} \max_{m \in \mathcal{M}} \sum_{n \neq m} \frac{c_n}{c_m} \|A_{\{m,n\}} F_{\{m,n\}}\|_1, \quad (18)$$

subject to $R_e F_e = k_e$ on every edge.

Theorem 5.5 (Fixed-weight Hermitian strong dual). *For $\alpha = (\alpha_m)_{m \in \mathcal{M}}$ in the probability simplex, put*

$$\omega_{mn}(\alpha, c) = \alpha_m \frac{c_n}{c_m} + \alpha_n \frac{c_m}{c_n} \quad (m < n).$$

Then

$$\begin{aligned} \Theta_c(\mathbf{k}) = \max \quad & \sum_{m < n} [\lambda_{mn}, k_{mn}]_{\mathbb{R}} \\ \text{subject to} \quad & \alpha_m \geq 0, \quad \sum_m \alpha_m = 1, \\ & A_{mn}^* z_{mn} = R_{mn}^* \lambda_{mn}, \quad \|z_{mn}\|_{\infty} \leq \omega_{mn}(\alpha, c) \quad (m < n). \end{aligned} \quad (19)$$

Both sides are attained, and

$$\Theta_c(\mathbf{k}) = \max_m \sum_{n \neq m} \frac{c_n}{c_m} \gamma_{mn}. \quad (20)$$

Proof. Introduce an epigraph variable η and one inequality

$$\sum_{n \neq m} \frac{c_n}{c_m} \|A_{mn} F_{mn}\|_1 \leq \eta$$

for every row m . Give these inequalities multipliers $\alpha_m \geq 0$. The infimum in η is finite exactly when $\sum_m \alpha_m = 1$. Edge $\{m, n\}$ then carries the norm multiplier $\omega_{mn}(\alpha, c)$. Applying [theorem 3.1](#) edge by edge gives (19). Strict feasibility of the row inequalities after choosing any feasible completions gives strong duality and dual attainment.

This also covers $\omega_{mn} = 0$: under the real pairing,

$$(\omega \|A \cdot\|_1)^*(\xi) = \begin{cases} 0, & \exists z : A^* z = \xi, \|z\|_{\infty} \leq \omega, \\ +\infty, & \text{otherwise.} \end{cases}$$

For (20), every edge cost is at least γ_{mn} , giving the lower bound row by row. Simultaneously choose all edge minimizers to obtain equality. $\square$

Corollary 5.6 (Collatz reduction).

$$\rho(Q^\gamma) = \inf_{c > 0} \Theta_c(\mathbf{k}) = \inf_{c > 0} \max_m \sum_{n \neq m} \frac{c_n}{c_m} \gamma_{mn}.$$

The infimum is valid for reducible Q^γ , although a strictly positive minimizing c need not exist.

Proof. This is the reducible-safe Collatz–Wielandt formula for the symmetric nonnegative matrix Q^γ . $\square$

Remark 5.7 (Do not dualize through the c -infimum). [Theorem 5.5](#) is a convex dual for each fixed positive c . The ratios c_n/c_m make joint optimization in $(c, \mathbf{F})$ nonconvex in these coordinates. Taking the infimum of the fixed- c primal values is legitimate; claiming a single strong dual after moving that infimum through the dual maximization is not.

6 Shared affine completion rules

The independent-edge optimum is the broadest analytic benchmark. A physical construction may require one declared rule across many row pairs. We now give the exact finite dual after such a rule is imposed.

6.1 Global constraint operator

Let

$$\mathcal{X}_{\mathcal{E}} = \bigoplus_{e \in \mathcal{E}} \mathcal{X}_e$$

contain one upper-triangle kernel per unordered edge. A declared affine rule is encoded by a real-linear map

$$\mathcal{R} : \mathcal{X}_{\mathcal{E}} \longrightarrow \mathcal{K}_{\text{glob}}$$

and a target $\kappa \in \mathcal{K}_{\text{glob}}$. The components of $\mathcal{R}\mathbf{F} = \kappa$ include every primitive restriction $R_e F_e = k_e$ and may also include:

- equality of shared interpolation parameters;
- a fixed q -resolved content-lift relation;
- block, side, or scale compatibility;
- a linear parametrization shared by one explicitly stacked finite coefficient family.

Nonlinear or data-adaptive rules are not silently included.

Remark 6.1 (Fixed data versus a coefficient class). [Theorem 6.2](#) below fixes one finite target κ and one positive Schur weight c . A rule chosen before a growing or infinite coefficient class introduces an additional $\inf_{\text{rule}} \sup_{\text{coefficients}}$ quantifier. It is covered only when the relevant finite coefficient family is explicitly stacked inside κ and $\mathcal{R}$; it is not a consequence of the single-datum theorem.

For fixed $c > 0$, define

$$\begin{aligned} \Theta_c^{\mathcal{R}}(\kappa) &= \min_{\eta, \mathbf{F}} \quad \eta \\ \text{subject to} \quad & \mathcal{R}\mathbf{F} = \kappa, \\ & \sum_{n \neq m} \frac{c_n}{c_m} \|A_{\{m,n\}} F_{\{m,n\}}\|_1 \leq \eta \quad (m \in \mathcal{M}). \end{aligned} \tag{21}$$

Theorem 6.2 (Coupled-rule strong dual). *If $\kappa \in \text{Ran } \mathcal{R}$, then*

$$\begin{aligned} \Theta_c^{\mathcal{R}}(\kappa) &= \max \quad [\lambda, \kappa]_{\mathbb{R}} \\ \text{subject to} \quad & \alpha_m \geq 0, \quad \sum_m \alpha_m = 1, \\ & \bigoplus_{m < n} A_{mn}^* z_{mn} = \mathcal{R}^* \lambda, \\ & \|z_{mn}\|_{\infty} \leq \alpha_m \frac{c_n}{c_m} + \alpha_n \frac{c_m}{c_n} \quad (m < n). \end{aligned} \tag{22}$$

The primal and dual optima are attained. If $\kappa \notin \text{Ran } \mathcal{R}$, the primal is infeasible and the dual is unbounded above.

Proof. Use row multipliers α_m exactly as in [theorem 5.5](#), and use λ for the single global equality constraint. The infimum in η imposes $\sum_m \alpha_m = 1$. The support-function representation of every edge norm introduces z_{mn} with the displayed radius. The infimum in the global variable $\mathbf{F}$ is finite precisely when the displayed block-adjoint stationarity equation holds. A feasible $\mathbf{F}$ and a strictly larger η give relative Slater feasibility, so finite-dimensional convex duality gives equality and dual attainment.

Primal attainment follows separately. Fix one feasible $\mathbf{F}_0$, quotient $\ker \mathcal{R}$ by the common zero-cost subspace

$$\{\mathbf{G} \in \ker \mathcal{R} : A_e G_e = 0 \text{ for every } e\},$$

and choose a linear complement. On that finite-dimensional complement, the maximum of the positively weighted row norms controls the variable, so a minimizing sequence has a convergent representative. For dual attainment, α lies in a compact simplex and every z_e is uniformly bounded. Restricting λ to $(\ker \mathcal{R}^*)^{\perp}$ then makes the adjoint stationarity equation bound λ , and the feasible set is closed.

The infeasible case follows by separating κ from $\text{Ran } \mathcal{R}$, as in [theorem 3.1](#). $\square$

Corollary 6.3 (Exact shared-rule obstruction). *Every feasible tuple $(\alpha, \lambda, (z_{mn}))$ in (22) proves that every completion family obeying the declared shared rule has weighted-Schur cost at least*

$$[\lambda, \kappa]_{\mathbb{R}}.$$

An optimal tuple proves the best possible lower obstruction for that rule and that fixed c .

Remark 6.4 (Restriction of scope). A witness for one $\mathcal{R}$ does not obstruct a different completion class. Conversely, the independent-edge value can be strictly smaller than the shared-rule value because its feasible set is larger. Any negative conclusion must name the exact rule class that its witness covers.

6.2 Hermitian symmetrization

Suppose both orientations are initially represented by legal completions. Define

$$\widehat{F}_{mn}(u, v) = \frac{1}{2} \left(F_{mn}(u, v) + \overline{F_{nm}(v, u)} \right)$$

and reconstruct the reverse by conjugate transpose.

Proposition 6.5 (No factor loss from reversal). *If the two orientation costs are transpose compatible, then*

$$\mathcal{C}_D(\widehat{F}_{mn}) \leq \frac{1}{2} (\mathcal{C}_D(F_{mn}) + \mathcal{C}_D(F_{nm})).$$

The symmetrized pair preserves the common Hermitian primitive datum.

Proof. The primitive restrictions agree by the inherited reversal identity. The cost is the ℓ^1 -norm of a linear analysis map, so convexity and transpose compatibility give the bound. $\square$

Remark 6.6. The proposition confirms that Hermitian compatibility itself costs no fixed factor. Shared q -provenance or a finite stacked-data rule may still impose a genuine cross-edge cost because those are additional constraints, not consequences of reversal.

7 Countermodels and non-substitution laws

The following finite examples separate the exact conclusions of the duality theory from the physical estimates that remain open.

7.1 A shared rule creates a real joint gap

Let one scalar $t \in \mathbb{R}$ control two edge completions, with costs

$$p_{12}(t) = |t|, \quad p_{23}(t) = |1 - t|.$$

Each edge separately has optimal cost zero. Under the shared rule,

$$\inf_t \max\{p_{12}(t), p_{23}(t)\} = \frac{1}{2}.$$

Thus a joint gap can be real, but it comes from the common parameter, not from Hermitian reversal.

7.2 Adaptive choice reverses max and min

In the same example,

$$\max_{e \in \{12, 23\}} \inf_t p_e(t) = 0, \quad \inf_t \max_{e \in \{12, 23\}} p_e(t) = \frac{1}{2}.$$

Choosing the completion after the worst edge or operator extremizer has been revealed changes the problem. A coefficient-uniform theorem must use the second order of quantifiers.

7.3 Fixed-pair decay does not control a matrix

Let P_N be the symmetric zero-diagonal matrix with

$$(P_N)_{mn} = \frac{1}{N-1} \quad (m \neq n).$$

Every fixed entry tends to zero, but

$$\rho(P_N) = 1.$$

Hence even optimal entrywise completion costs of order $o(1)$ are insufficient without a row, weighted-Schur, or Perron aggregation over the complete growing active set.

7.4 The absolute Perron floor can miss signed cancellation

Let $N = 2^r$, and let S_N be the symmetric Sylvester Hadamard matrix. Then

$$S_N = S_N^T, \quad S_N S_N^T = NI, \quad |(S_N)_{mn}| = 1.$$

Define the real symmetric zero-diagonal matrix

$$H_N = \frac{S_N - \text{diag}(S_N)}{N}.$$

Since $\|S_N\| = \sqrt{N}$,

$$\|H_N\| \leq \frac{1}{\sqrt{N}} + \frac{1}{N} \rightarrow 0.$$

On the other hand,

$$|H_N| = \frac{J - I}{N}, \quad \rho(|H_N|) = \frac{N-1}{N} \rightarrow 1.$$

By [theorem 5.3](#), every entrywise absolute completion majorant of this signed matrix has spectral radius asymptotically at least one. Thus failure of the whole completion–Perron architecture need not be failure of signed operator cancellation.

7.5 One bad completion is not an obstruction

Take zero primitive datum, a prime $p > D$, and a rectangle containing the nonprimitive site (p, p) . The fill

$$F_A = Ae_{p,p}$$

is legal for every $A \in \mathbb{C}$. The tail coordinate $(d, x, y) = (p, 1, 1)$ has unit weight, so its exact tail cost is at least $|A|$. The optimum remains zero because the zero completion is legal. A large cost for a chosen interpolation proves nothing about $\gamma_D(0)$; only an optimal dual witness can lower-bound the infimum.

7.6 A dual obstruction is not a signed lower bound

In the abstract normal form $R = I$, $A = I$, take

$$k = (1, -1), \quad \sigma = (1, 1).$$

Then the signed functional vanishes:

$$\langle \sigma, k \rangle = 0,$$

whereas the completion gauge is

$$\gamma(k) = \|k\|_1 = 2.$$

The optimal dual witness obstructs a cheap ℓ^1 certificate, not the signed functional. The Hadamard construction above is the matrix version of the same separation.

7.7 Costs before content aggregation can discard cancellation

Suppose two literal content slices satisfy

$$L_{mn,q_1} = F, \quad L_{mn,q_2} = -F.$$

The physically relevant post- q -aggregation datum is zero. Completing the two slices separately and adding their absolute costs can be positive, whereas completing their exact sum costs zero. Therefore a paper must declare whether its completion rule acts after exact q -aggregation or retains q as a coupled label. One may not move between the two orders after seeing which is cheaper.

7.8 Sidewise renormalization changes the object

Suppose two orthogonal sides each contribute row mass one, so the raw global masses are $W_m = W_n = 2$. If one side contains one common unit key, its correct normalized two-corner contribution is

$$\frac{1}{\sqrt{W_m W_n}} = \frac{1}{2}.$$

Renormalizing that side alone gives 1, doubling the physical entry. Analytic fill values never enter W_m or W_n ; a completion cannot authorize a new normalization.

7.9 Hermitian-incompatible scalar completions

For zero primitive data, both

$$F_{12} = A e_{2,2}, \quad F_{21} = 0$$

are legal scalar completions. Unless $A = 0$, they do not satisfy

$$F_{21}(v, u) = \overline{F_{12}(u, v)}.$$

They cannot be inserted together into one Hermitian analytic carrier. One must use upper-triangle variables or apply [proposition 6.5](#).

7.10 A fixed weight is not automatically Perron-optimal

For a nonnegative matrix P and an arbitrary $c > 0$,

$$\rho(P) \leq \max_m \frac{(Pc)_m}{c_m},$$

and the inequality may be strict. Equality requires an appropriate Perron weight or the infimum over all positive c . A dual lower witness for one fixed- c Schur problem does not by itself lower-bound the infimum over c .

8 The literal fixed-shift completion gate

8.1 Inherited carrier

We now specialize only to the physical object proved in TPC-74 [5]. Fix one nonzero shift h_0 , one external block, and the actual TPC-61/TPC-68 missing-high carrier. The two direct sides are combined through their proved orthogonal direct sum. Every key retains its side, block, opposite row, profile, time, mask, endpoint, and remaining native output labels.

For an active unordered row pair $m < n$, write

$$\Delta_{mn} = h_0(n - m)$$

and let $L_{mn,q}(u, v)$ be the literal q -resolved direct-entry kernel of TPC-74. Its post-content aggregation is

$$L_{mn}(u, v) = \sum_{q \mid |\Delta_{mn}|} L_{mn,q}(u, v).$$

Before optimizing, choose a finite rectangle B_{mn} containing the union of the primitive supports of all $L_{mn,q}$ and every declared lifted nonprimitive site. Fix $D_{mn} \geq 1$, and set

$$B_{nm} = B_{mn}^\top, \quad D_{nm} = D_{mn}.$$

These choices are used in γ_{mn} , Q^γ , P^0 , and P^{lift} ; changing them changes the optimization problem. Prescribe

$$k_{mn}(u, v) = L_{mn}(u, v) \quad ((u, v) = 1), \quad (23)$$

including zero at every absent primitive site. Only nonprimitive sites are completion variables.

Here γ_{mn} is the pair-dependent optimum formed after exact q -aggregation of this fixed literal datum. In contrast, F_{mn}^{lift} below is a candidate informed by retained pre-aggregation q -provenance. It is not a slice-by-slice absolute-value optimization.

The normalization inside every $L_{mn,q}$ remains

$$\frac{1}{\sqrt{W_M(m)W_M(n)}}, \quad W_M(m) = \sum_{w:m(w)=m} |\beta_w|^2,$$

where the sum uses both orthogonal sides on the declared block. This is the W_m notation used in TPC-74. Nonprimitive fill values are not physical atoms and never enter W_M .

8.2 Three exact majorants

Let F_{mn}^0 be the canonical zero completion. Let γ_{mn} be the exact unrestricted analytic optimum from equation (4). TPC-74 and theorem 5.3 give

$$\|H\| = \|\tilde{H}\| \leq \rho(|\tilde{H}|) \leq \rho(Q^\gamma) \leq \rho(P^0). \quad (24)$$

Every matrix in this chain refers to the same literal row set, complete keys, raw masses, and fixed h_0 .

There is also a canonical provenance-preserving candidate that was not available before the q -resolved direct-entry representation. Write a site uniquely as

$$(x, y) = (du, dv), \quad d = (x, y), \quad (u, v) = 1.$$

Define the content lift, inside B_{mn} , by

$$F_{mn}^{\text{lift}}(du, dv) = \sum_{\substack{r||\Delta_{mn}| \\ d|r}} L_{mn,r}(u, v). \quad (25)$$

The sum is zero when no such r exists.

Proposition 8.1 (Literal content lift). *The kernel F_{mn}^{lift} is a legal completion of (23). It uses only actual q -slices with $q \mid |\Delta_{mn}|$, introduces no physical mass, and respects Hermitian reversal:*

$$F_{nm}^{\text{lift}}(v, u) = \overline{F_{mn}^{\text{lift}}(u, v)}.$$

Proof. At a primitive site $d = 1$, (25) gives

$$F_{mn}^{\text{lift}}(u, v) = \sum_{r||\Delta_{mn}|} L_{mn,r}(u, v) = k_{mn}(u, v).$$

Thus the primitive restriction is correct, including absent-ray zeros. Every nonprimitive value is only a sum of already existing literal content slices. It is not inserted into the physical key set or row mass, and (25) is a declared analytic copying rule, not a physical movement or identity between atoms. Finally, TPC-74 gives

$$L_{nm,r}(v, u) = \overline{L_{mn,r}(u, v)};$$

the divisor set of $|\Delta_{mn}|$ is unchanged by reversal. $\square$

Let P^{lift} be the symmetric matrix of the full costs $\mathcal{C}_{D_{mn}}(F_{mn}^{\text{lift}})$. Then

$$\rho(Q^\gamma) \leq \min\{\rho(P^0), \rho(P^{\text{lift}})\}. \quad (26)$$

No ordering between P^0 and P^{lift} is asserted. The lift may reduce the $d = 1$ plaquette term while increasing higher diagonal terms or the tail.

8.3 Exact positive and negative outcomes

Let $X \rightarrow \infty$ through one declared physical family, with all objects above indexed by X .

Positive separated outcome. If one proves

$$\rho(Q_X^\gamma) \leq \eta_X < 1,$$

then the normalized missing-native Gram

$$G_{\text{nat},X} = I + H_X$$

obeys

$$U_{d,X} G_{\text{nat},X} U_{d,X} = I + \tilde{H}_X.$$

The two Grams are unitarily equivalent, so

$$1 - \eta_X \leq \lambda_{\min}(G_{\text{nat},X}) \leq \lambda_{\max}(G_{\text{nat},X}) \leq 1 + \eta_X.$$

This is a deterministic implication. The hypothesis is not proved here.

Negative separated outcome. For each edge, a dual witness gives the nonnegative lower bound

$$W_{mn} = |[\lambda_{mn}, k_{mn}]_{\mathbb{R}}| \leq \gamma_{mn};$$

changing the common sign of (λ_{mn}, z_{mn}) makes this absolute value a feasible dual objective. Set $W_{nm} = W_{mn}$ and $W_{mm} = 0$. Then

$$\rho(W) \leq \rho(Q^\gamma).$$

Thus a nonnegligible actual-carrier lower bound for $\rho(W)$ closes the broad independent absolute-completion route at that scale. It still does not lower-bound $\|H\|$.

Shared-rule outcome. For a declared affine q -compatible rule at one fixed finite stacked datum, [theorem 6.2](#) gives the exact fixed- c value and its obstruction. A witness for one c does not close the Perron infimum; one needs a bound uniform over positive c , a direct nonnegative matrix floor, or an exact evaluation of the resulting Collatz problem. Nor does the single-datum theorem supply a rule uniform over a growing coefficient class; that requires the additional minimax quantifier in [remark 6.1](#).

8.4 Remaining physical gates

Even a successful missing-native estimate must still be combined on one carrier with:

- (i) terminal activation and the represented lower frame;
- (ii) missing-mass comparison and nuisance shorting;
- (iii) geometric endpoint and packet-boundary terms; and
- (iv) the complete fixed-power loss ledger.

If the accumulated loss reaches $1/400$, including equality, the current strict endpoint implication stops. None of the completion identities changes that budget.

8.5 TPC–76 decision

TPC–75 leaves two rigorously separated next doors.

- (a) Analyze the literal content lift [\(25\)](#) on the divisor poset and compare its exact KKT/dual flow with the zero completion.
- (b) If an actual absolute-Perron floor is already nonnegligible, pivot to a signed full- H moment or a hybrid direct/anchor architecture, because no entrywise absolute completion can cross that floor.

The first route is the default theoretical next step; the second is an explicit kill-criterion fallback rather than an overclaim that cancellation has failed.

9 Claim ledger and conclusion

9.1 Exact conclusions

This paper proves:

- (1) an exact finite linear analysis operator whose ℓ^1 -norm is the complete divisor–variation completion cost;
- (2) primal and dual attainment, with no duality gap, for every feasible finite complex completion problem, together with the exact infeasible/unbounded alternative;
- (3) the quotient-seminorm identity $\ker \gamma = R(\ker A)$;
- (4) exact dual obstruction witnesses and the arithmetic Möbius witness as one feasible dual direction;
- (5) the factorization $\inf_{\mathbf{F}} \rho(P(\mathbf{F})) = \rho(Q^\gamma)$ for independent unordered row pairs;
- (6) the universal completion sandwich

$$\|H\| = \|\tilde{H}\| \leq \rho(|\tilde{H}|) \leq \rho(Q^\gamma) \leq \rho(P^0);$$

- (7) a fixed-weight Hermitian row-Schur strong dual;
- (8) a strong dual for every declared finite affine shared rule at fixed stacked data and fixed positive Schur weight;
- (9) a literal q -provenance content lift for the TPC–74 direct kernels; and
- (10) deterministic countermodels separating certificate failure from signed cancellation failure.

9.2 Statements not proved

The paper does not prove:

- that γ_{mn} is small relative to the actual raw entry mass on a growing physical family;
- that the content lift is uniformly cheaper than the canonical zero completion;
- a row-sum, weighted-Schur, or Perron saving for Q_X^γ ;
- that $\rho(|H_X|) = o(1)$, or any fixed-power decay;
- that a finite numerical SOCP certificate persists asymptotically;
- that an obstruction for one shared rule covers all completion rules;
- cancellation in the same-residual sector or the full signed matrix;
- closure of the represented frame, missing-mass comparison, nuisance shorting, endpoints, or final boundary reconstruction;
- the strict $1/400$ endpoint inequality;
- a result special to $h_0 = 2$; or
- a parity-breaking, prime-pair, or twin-prime theorem.

9.3 Audit levels

Result	Scope	Level
Analysis normal form, strong duality, quotient gauge	All feasible finite linear ℓ^1 systems, with infeasibility alternative	L0
Independent-edge Perron factorization	All finite Hermitian-compatible product classes	L0
Affine shared-rule dual	Every fixed-data finite real-linear coupling	L0
Hadamard Perron-floor separation	Finite symmetric matrices of order 2^r	L0
Direct-entry gauge and content lift	Complete literal TPC-74 missing-high carrier	L1
Normalized completion saving	Complete growing physical row-pair family	Open L2
Perron decay and physical re-assembly	One common carrier and coefficient scope	Open

9.4 Conclusion

The analytic completion gate is no longer an open-ended interpolation problem. At every declared finite scale,

$$\min_{RF=k} \|A_DF\|_1 = \max_{\substack{A_D^* z = R^* \lambda \\ \|z\|_\infty \leq 1}} \operatorname{Re}\langle \lambda, k \rangle.$$

A construction and an obstruction are exact primal-dual certificates for the same number. On the complete TPC-74 direct-entry carrier, the independent optimal values form one symmetric matrix Q^γ , and its Perron radius is exactly the best separated completion majorant.

This closes two conceptual gaps. First, a failed completion search can now be replaced by a rigorous obstruction rather than a heuristic negative claim. Second, Hermitian reversal is not blamed for a joint gap it does not create; any such gap must be traced to an explicitly shared rule. The Hadamard separation then supplies the correct stop rule: a large absolute-Perron obstruction closes the completion certificate, but it leaves a signed-moment route open.

The next task is therefore concrete. Analyze the literal divisor-poset content lift and its dual flow on the actual carrier. If the actual-carrier absolute Perron floor is already nonnegligible, stop that route and change architecture. If the lift improves the complete row matrix above that unavoidable floor, the result is immediately compatible with the same raw normalization and fixed-shift operator chain.

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